CE 4011 Static + Modal Desktop MVP User Manual

Final Documentation Package

Spring Semester 2025–2026

1 Final Desktop Scope

The desktop MVP supports two-dimensional Static Analysis and Modal Analysis. Use the Tkinter application to define or load a model, validate it, solve it, inspect educational tables and plots, and export visible results where implemented. RSA and THA are retained only as future-extension backend work and are not part of the submitted desktop workflow.

2 Overall Workflow

The normal desktop workflow is to start from a blank model, the 2D shear-frame template, or an XML file; inspect and edit the model; validate it; run either Static Analysis or Modal Analysis; then review and export the visible results. The diagram source is `desktop_workflow.puml` in the PlantUML diagram folder.

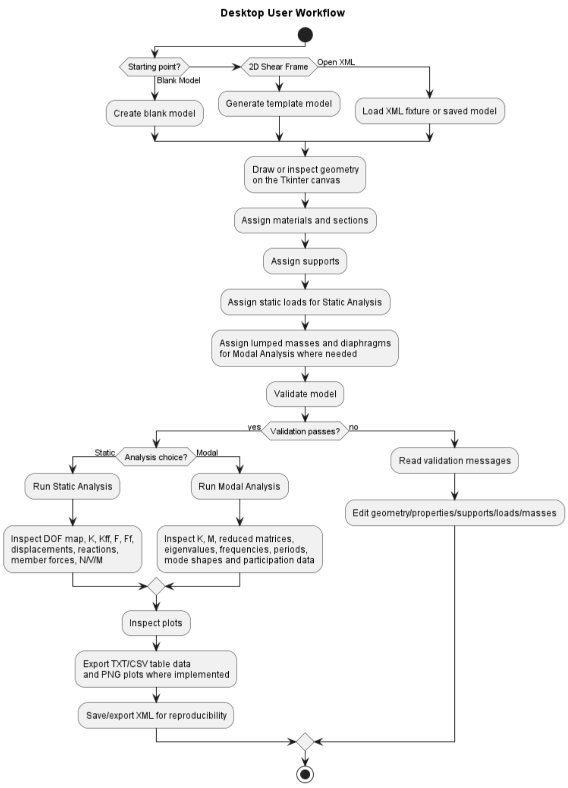

Figure 1: Desktop workflow

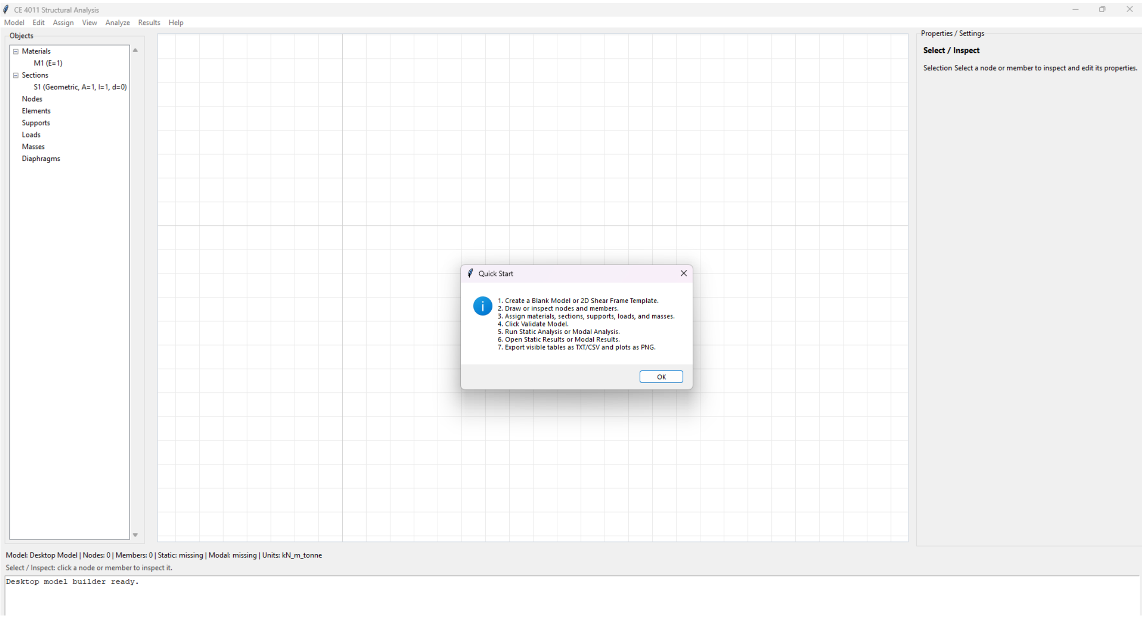

Figure 2: Help then Quick Start

3 Starting a Model

3.1 Blank Model

Choose a blank model when entering a custom topology. Define nodes by coordinates, then connect them with elements. This workflow is appropriate for verification examples, trusses, frames, cantilevers, and small teaching models.

Blank model workflow

Screenshot placeholder for creating a blank model.

TODO: Insert final screenshot.

Figure 3: Blank model workflow

3.2 2D Shear Frame Template

Choose the two-dimensional shear-frame template when the model is a regular frame with repeated bays or stories. Enter the requested geometry and property data, then review the generated nodes and elements before assigning final loads or masses.

4 Model Editing

4.1 Geometry and Canvas

Use the model canvas to inspect node and element layout. The canvas is a visual editing surface; the numerical model is still passed through `ModelBuilder` and `StructuralModel`.

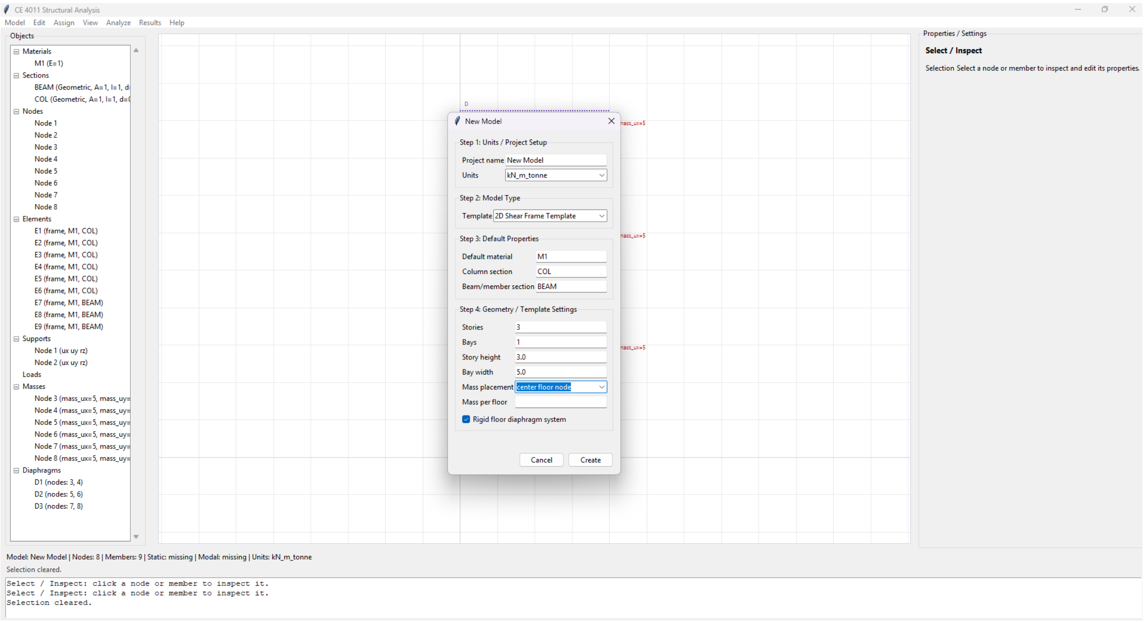

Figure 4: 2D shear frame template

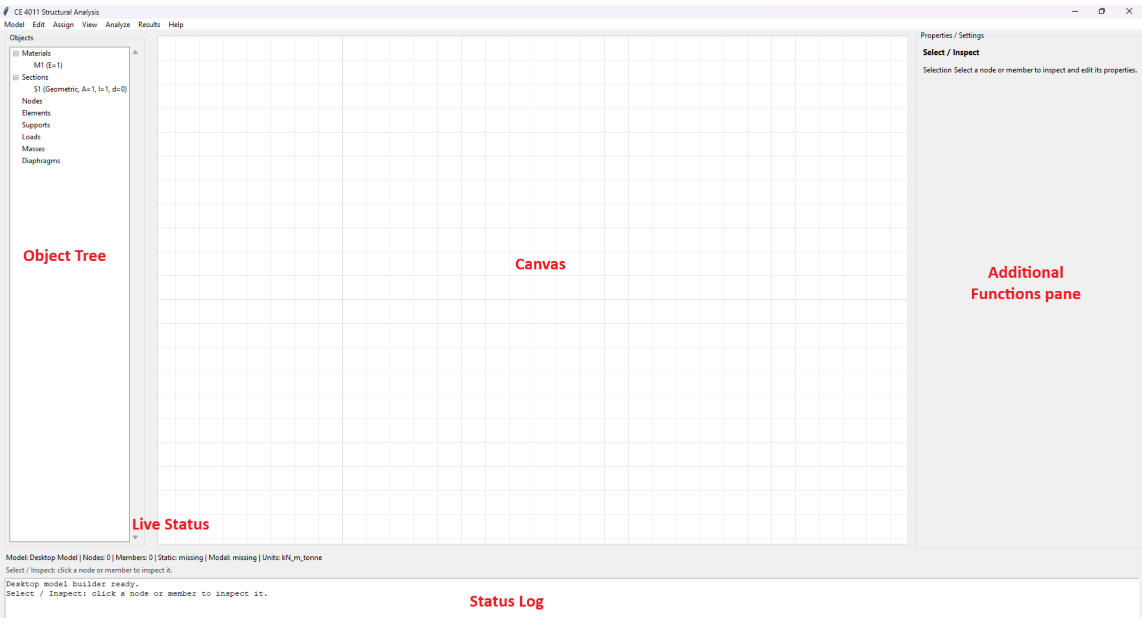

Figure 5: Model canvas

4.2 Property Panel

Select model objects to review or edit their properties. The property panel should be used for coordinates, labels, element connectivity, support values, load values, mass values, and related object data.

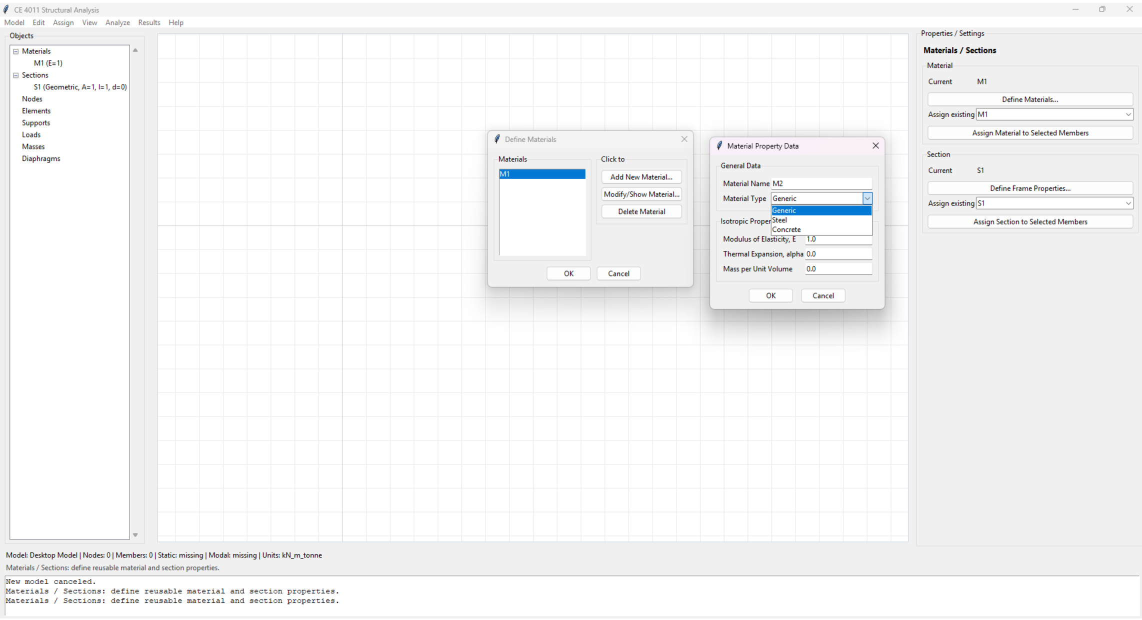

Figure 6: Property panel

4.3 Materials and Sections

Define material properties such as elastic modulus and section properties such as area and moment of inertia before assigning them to elements. These values control element stiffness and, where density is modeled, modal mass contribution.

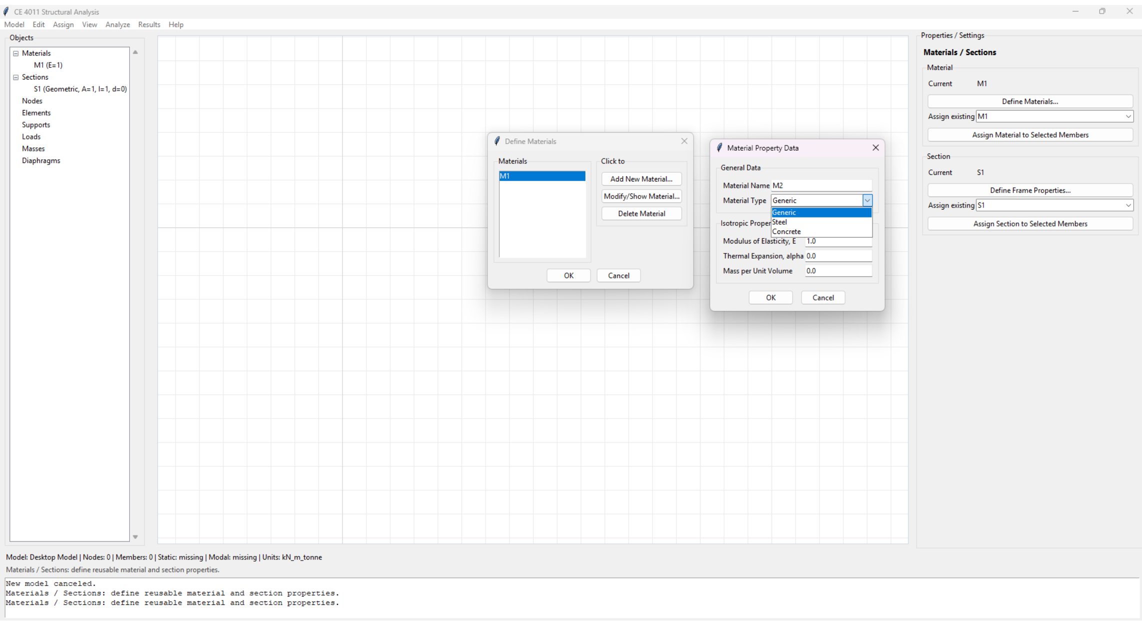

Figure 7: Materials/sections assignment

4.4 Supports, Loads, Settlements, Masses, and Diaphragms

Assign supports before solving so the model has stable boundary conditions. Assign loads for Static Analysis. Assign nodal or element mass data for Modal Analysis. Where supported by the model input, assign support settlements and diaphragm data as needed.

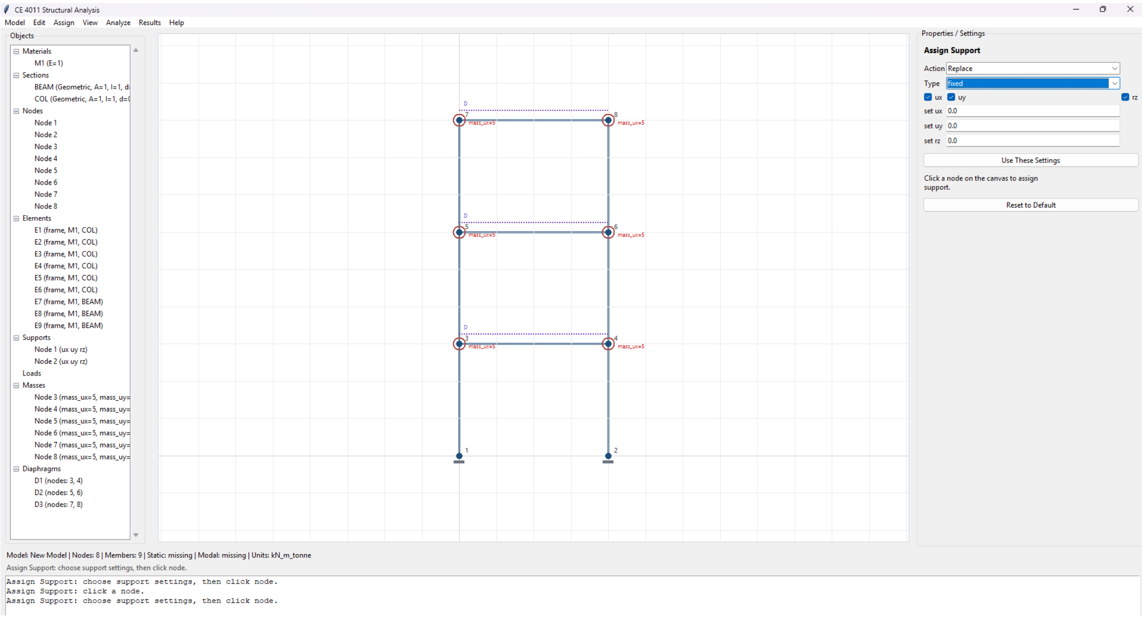

Figure 8: Supports assignment

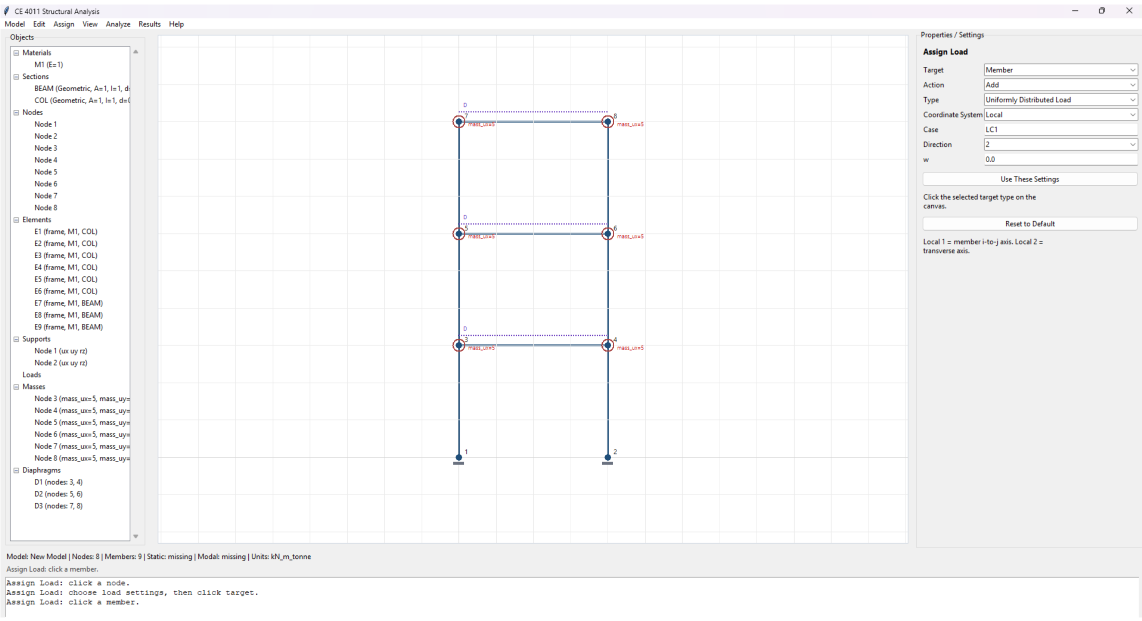

Figure 9: Member load assignment

5 Validation

Run model validation before analysis. Validation should catch missing properties, unstable support conditions, invalid connectivity, and other model definition errors that would prevent meaningful analysis.

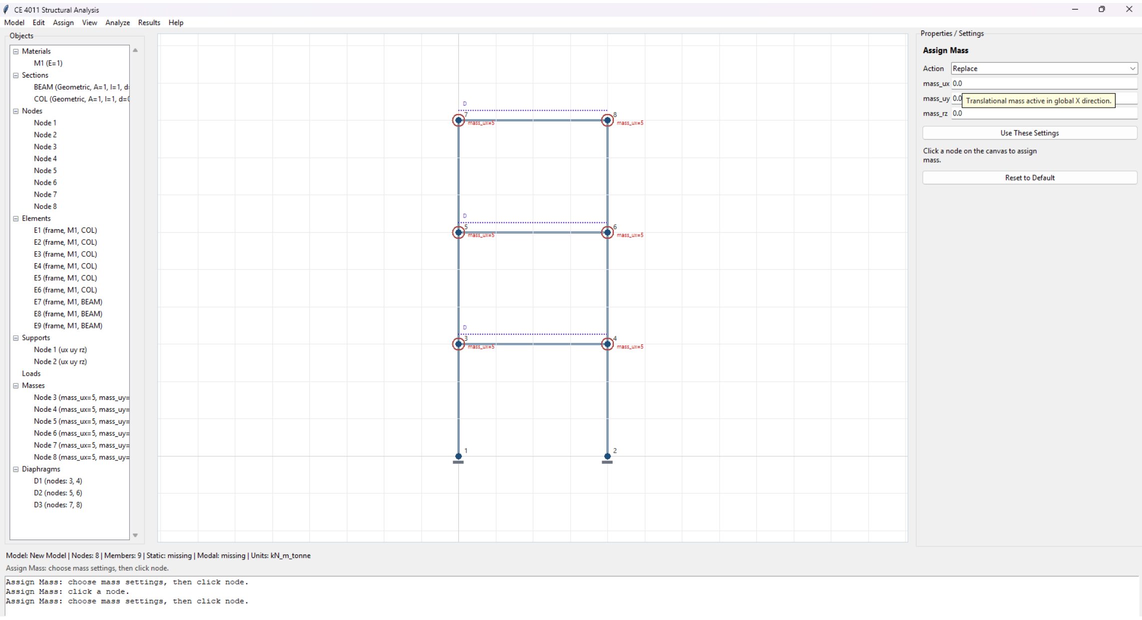

Figure 10: Mass assignment

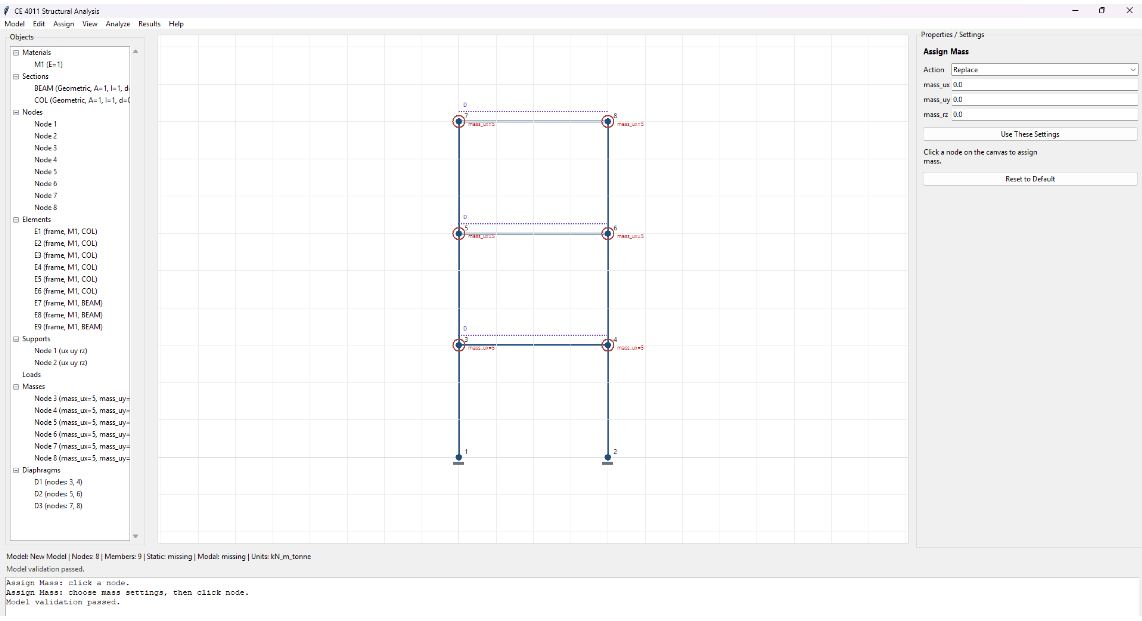

Figure 11: Validation status/dialog

6 Running Static Analysis

Run Static Analysis after defining geometry, element properties, boundary conditions, and loads. The solver assembles the full stiffness matrix K, global force vector F, reduced free-DOF matrix K_{ff}, and reduced force vector F_f, then solves for nodal displacements and recovers reactions and member forces.

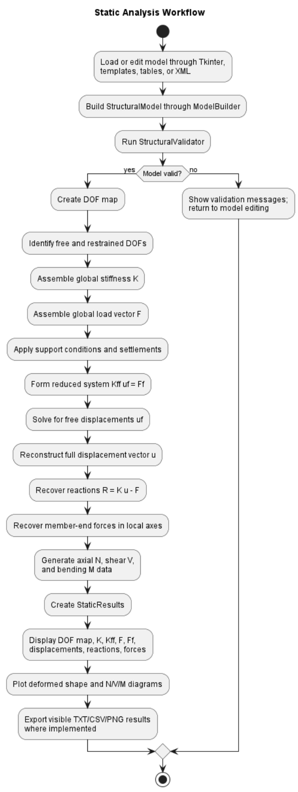

Figure 12: Static analysis workflow

6.1 Interpreting Static Tables

Table 1: Static result interpretation

Output	Meaning
DOF map	Maps each node component $[u_x, u_y, r_z]$ to a global equation number and shows which DOFs are free or restrained.
K	Full assembled global stiffness matrix before boundary reduction.
K_{ff}	Reduced stiffness matrix for free DOFs after applying support conditions.
F	Full assembled global force vector, including equivalent nodal actions where applicable.
F_f	Reduced force vector for free DOFs, adjusted for prescribed support displacements when applicable.
Nodal displacements	Solved translations and rotations at free DOFs, reconstructed into nodal form.

Output	Meaning
Support reactions	Resisting forces and moments recovered at restrained DOFs from $R = Ku - F$.
Member-end forces	Local element end forces recovered from element displacement and fixed-end effects.
N/V/M diagrams	Axial force, shear force, and bending moment data along each member using the project sign convention.

7 Running Modal Analysis

Run Modal Analysis after defining geometry, stiffness properties, boundary conditions, and mass data. The solver assembles stiffness and mass matrices, handles inactive and massless DOFs, solves the generalized eigenproblem, and reports modal properties.

7.1 Interpreting Modal Tables

Table 2: Modal result interpretation

Output	Meaning
Eigenvalues	Values $\lambda = \omega^2$ from $K\phi = \lambda M\phi$.
Frequencies	Cyclic frequencies $f = \omega/(2\pi)$.
Periods	Modal periods $T = 1/f$.
Mode shapes	Relative displacement patterns for each mode. Display scaling is for readability.
Modal mass	Generalized mass for each mode; mass-normalized modes have unit modal mass.
Participation factors	Directional participation values showing how strongly each mode contributes to the selected influence vector.
Effective modal mass	Equivalent participating mass represented by each mode.
Mass participation ratios	Effective modal mass divided by total participating mass; cumulative ratios show captured dynamic mass.

8 Exporting Results

Use result-window export controls to save visible result tables and plots where implemented. Static table TXT export, Modal table CSV export, and Static plot PNG export are included in the final desktop command-path validation. XML save/export is used to preserve or reproduce the model itself.

9 Complete Static Analysis Example

This example uses `data/test-frame.xml`, the retained static benchmark fixture, rather than an invented manual model. It is a four-node, 4 m by 3 m two-dimensional frame/braced-frame style model in `kN_m_tonne` units. It contains one material, two sections, four frame elements, supports at nodes 1 and 4, and load case `LC1` with point loads at nodes 2 and 3. The file does not define modal masses, so use it for Static Analysis.

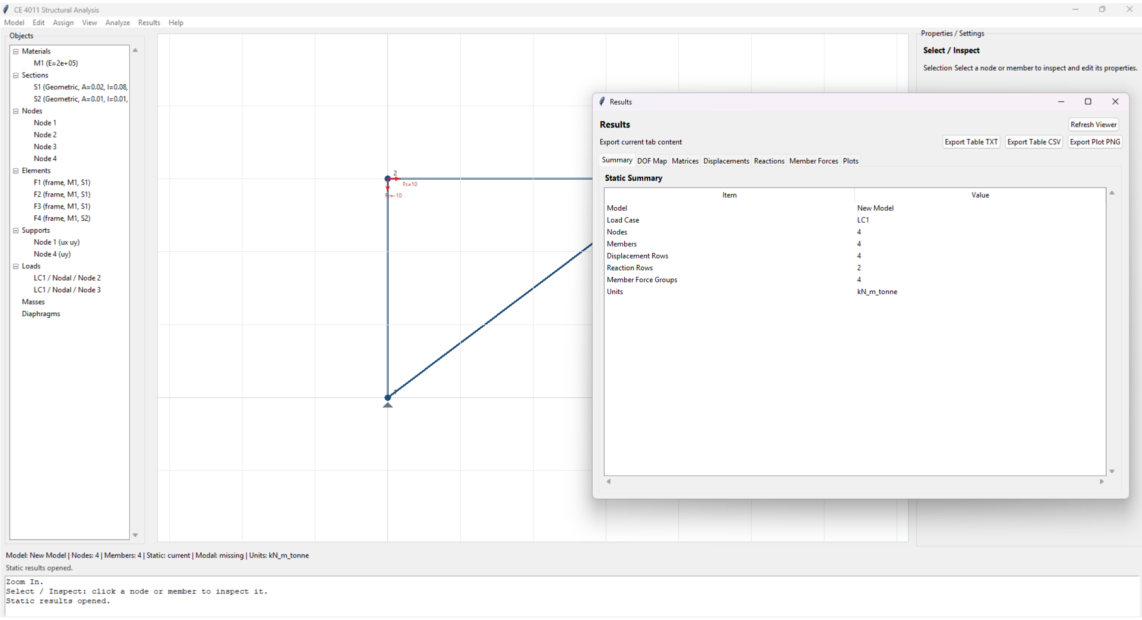

Figure 13: Static result summary

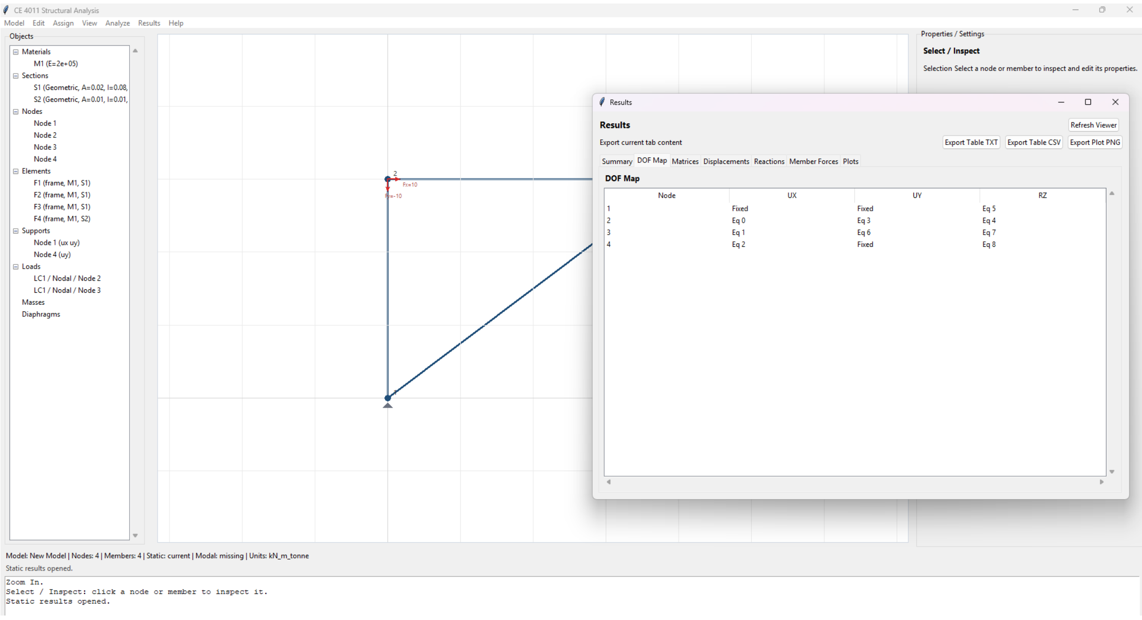

Figure 14: Static DOF map / K / Kff table

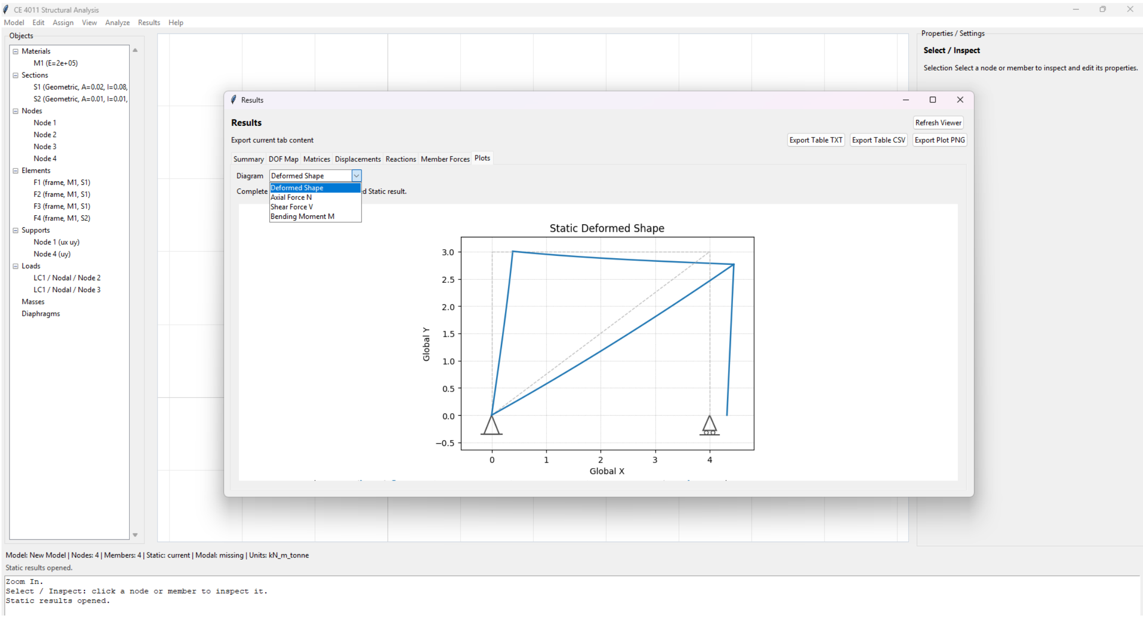

Figure 15: Static deformed shape

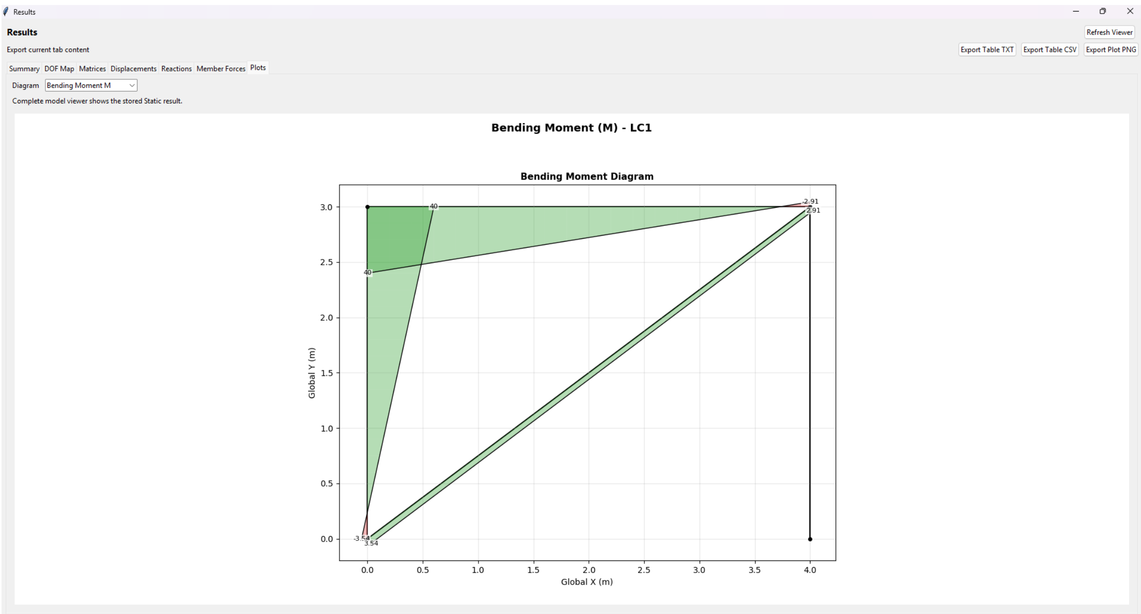

Figure 16: N/V/M diagram

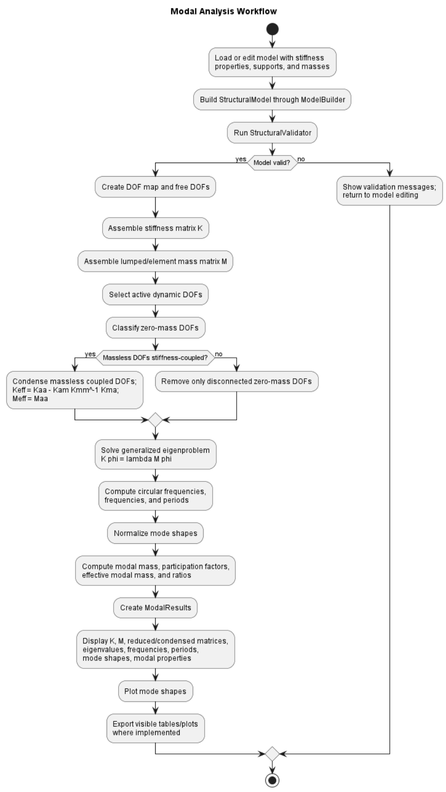

Figure 17: Modal analysis workflow

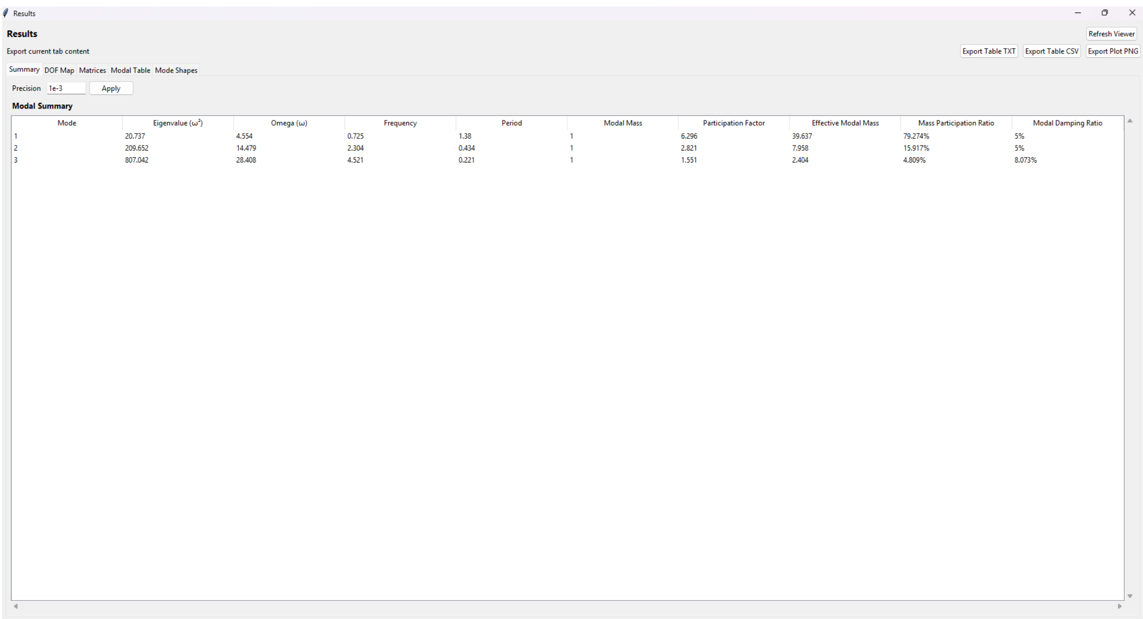

Figure 18: Modal result summary

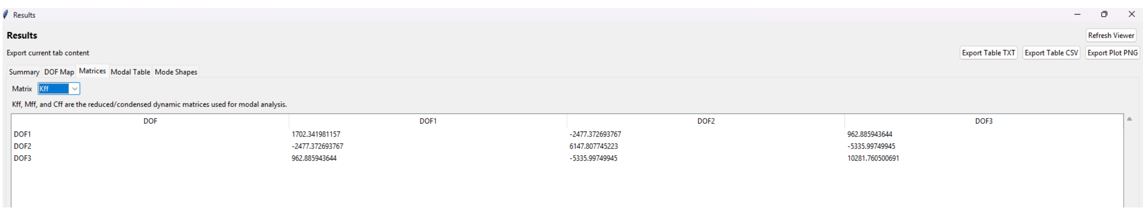

Figure 19: Modal K/M/reduced matrix view

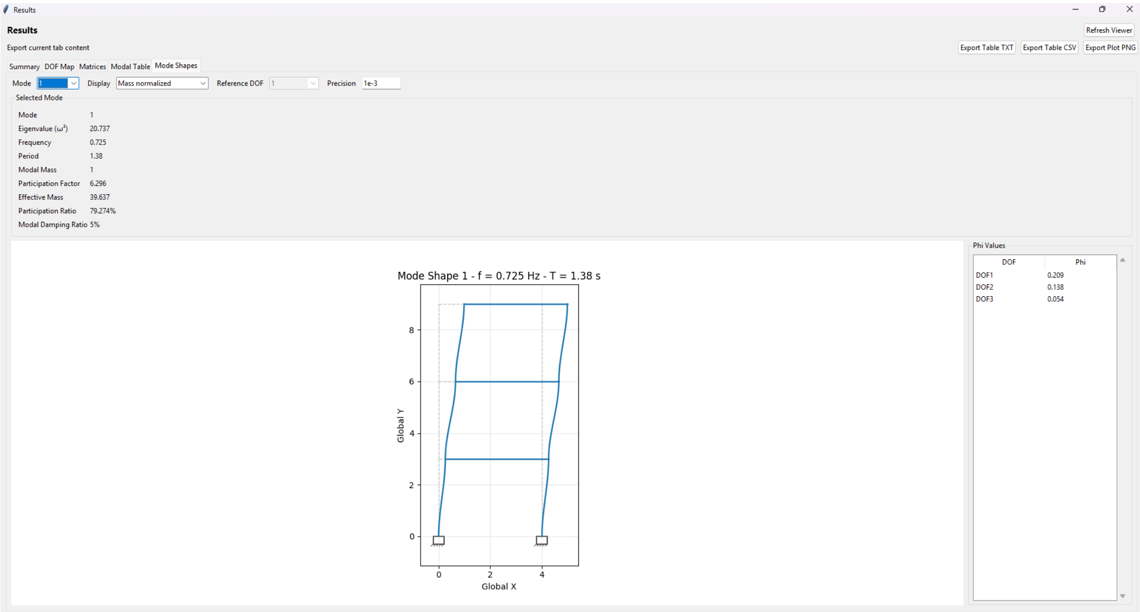

Figure 20: Mode shape plot

Table 3: Export files available for final documentation evidence

Export	Format	Evidence provided
Static table	TXT	Static export retained in `data/`. It records model `New Model`, load case `LC1`, four nodes, four members, four displacement rows, two reaction rows, four member-force groups, and `kN_m_tonne` units.
Static plots	PNG	Static deformed-shape and N/V/M plot figures are already placed in the LaTeX figure folder; no standalone static plot export file is present under `data/`.
Modal table	CSV	Modal export retained in `data/`. It records three modes, eigenvalues, circular frequencies, frequencies, periods, modal masses, participation factors, effective modal masses, mass participation ratios, and modal damping ratios.
Modal plots	PNG	Modal plot exports retained in `data/`. Figures used inside the LaTeX documents remain the images already placed under `docs/final_latex/`.

Export buttons

Screenshot placeholder for table and plot export buttons in result windows.

TODO: Insert final screenshot.

Figure 21: Export buttons

9.1 Static Fixture Summary

Table 4: Static example fixture

Item	Fixture content
XML file	`data/test-frame.xml`
Geometry	Four nodes: base nodes at $x = 0$ m and $x = 4$ m, upper nodes at $y = 3$ m.
Elements	Four frame elements: two vertical members, one top member, and one diagonal member.
Properties	Material `M1`; sections `S1` and `S2`.
Supports	Node 1 restrains u_x and u_y; node 4 restrains u_y. Rotations are not restrained in this fixture.
Loads	Load case `LC1` applies nodal point loads at nodes 2 and 3.
Analysis use	Static Analysis example and static benchmark documentation evidence.

9.2 Static Procedure

1. Launch the desktop app with `python -m src.ui_desktop.app`.
2. Choose `Model -> Open XML` and open `data/test-frame.xml` from the repository.
3. Inspect the geometry on the canvas and review the object tree/property panel to confirm nodes, elements, material, sections, supports, and load case `LC1`.
4. Choose `Analyze -> Validate Model`. Continue only after the validation message reports a valid model.
5. Choose `Analyze -> Run Static Analysis`.
6. Choose `Results -> Static Results`.
7. Inspect the `Summary`, `DOF Map`, `Matrices`, `Displacements`, `Reactions`, `Member Forces`, and `Plots` tabs.
8. In the `Plots` tab, review the deformed shape and the axial force N, shear force V, and bending moment M diagrams.
9. Export visible tables as TXT or CSV and export visible plots as PNG where needed for the report or appendix.
10. Use `Model -> Export XML` only when a copy of the reproducible model file is needed.

9.3 Static Screenshots Still To Insert

Insert screenshots after the final desktop run for the loaded canvas geometry, successful validation message, Static Results summary, DOF map and matrix views, displacement/reaction/member-force tables, deformed shape plot, N/V/M plots, and any TXT/CSV/PNG export evidence used in the report.

Static fixture output

TODO screenshot placeholder for `data/test-frame.xml`: loaded geometry, validation, Static Results summary, DOF/matrix tables, displacements/reactions/member forces, deformed shape, and N/V/M plots.

TODO: Insert final screenshot.

Figure 22: Static fixture output

10 Complete Modal Analysis Example

This example uses `data/test-modal-flex-beam.xml`, the retained modal flexible-beam/shear-frame benchmark fixture. It is a three-story, two-column frame with floor beams, fixed base supports at nodes 1 and 2, floor diaphragm groups, and horizontal lumped masses at the elevated floor nodes. The file has stiffness and mass data for Modal Analysis and no static load case, so use it for Modal Analysis.

10.1 Modal Fixture Summary

Table 5: Modal example fixture

Item	Fixture content
XML file	`data/test-modal-flex-beam.xml`
Geometry	Eight nodes arranged as two columns over three 3 m stories with 4 m bay width.
Elements	Six column elements and three floor-beam elements.
Properties	Material `M1`; direct section stiffness values through `EA` and `EI` on `BEAM`, `COL1`, and `COL2`.
Supports	Base nodes 1 and 2 are fixed in u_x, u_y, and r_z.
Mass model	Horizontal lumped masses at nodes 3 through 8; floor diaphragm groups tie floor-level horizontal motion.
Analysis use	Modal Analysis example and flexible-beam modal benchmark documentation evidence.

Table 6: Modal result values exported in `data/modal_results_table.csv`

Mode	λ	ω	f	T	M_n	Γ_n	Mass ratio
1	20.737	4.554	0.725	1.380	1.000	6.296	79.274%
2	209.652	14.479	2.304	0.434	1.000	2.821	15.917%
3	807.042	28.408	4.521	0.221	1.000	1.551	4.809%

10.2 Modal Procedure

1. Launch the desktop app with `python -m src.ui_desktop.app`.
2. Choose `Model -> Open XML` and open `data/test-modal-flex-beam.xml` from the repository.

3. Inspect the frame geometry on the canvas and review the property panel for the fixed base supports, floor diaphragms, and lumped masses.
4. Choose `Analyze -> Validate Model`. Continue only after the validation message reports a valid model.
5. Choose `Analyze -> Run Modal Analysis`. Use the modal settings dialog values selected for the final verification run.
6. Choose `Results -> Modal Results`.
7. Inspect the `Summary`, `DOF Map`, `Matrices`, `Modal Table`, and `Mode Shapes` tabs.
8. In the matrix views, confirm that stiffness and mass data are present and that reduced or condensed dynamic matrices are available where the solver reports them.
9. In the modal result views, inspect eigenvalues, frequencies, periods, mode-shape components, modal mass, participation factors, effective modal mass, and mass participation ratios.
10. Export visible modal tables as TXT or CSV and export selected mode-shape plots as PNG where needed for the report or appendix.

10.3 Modal Screenshots Still To Insert

Insert screenshots after the final desktop run for the loaded modal geometry, mass or diaphragm evidence, successful validation message, modal settings dialog, Modal Results summary, matrix view, Modal Table, selected Mode Shapes plot, and any TXT/CSV/PNG export evidence used in the report.

Modal fixture output

TODO screenshot placeholder for `data/test-modal-flex-beam.xml`: loaded geometry, validation, modal settings, Modal Results summary, matrices, Modal Table, Mode Shapes, and export evidence.

TODO: Insert final screenshot.

Figure 23: Modal fixture output

11 User Checklist

Before final submission or classroom demonstration, confirm:

- the model validates successfully;
- every element has material and section data;
- static models have loads and stable supports;
- modal models have appropriate mass data;
- result tables match the selected analysis type;
- exported tables/plots correspond to the visible results being discussed.